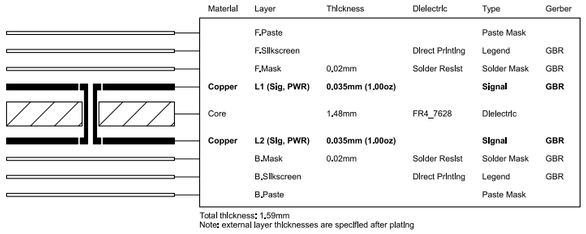
# BATTERY_INTERFACE_BOARD Fabrication Document

## Layer Stack Legend



| Material | Layer | Thickness | Dielectric | Type | Gerber |
|---|---|---|---|---|---|
| | F.Paste | | | Paste Mask | |
| | F.Silkscreen | | Direct Printing | Legend | GBR |
| | F.Mask | 0.02mm | Solder Resist | Solder Mask | GBR |
| **Copper** | **L1 (Sig, PWR)** | **0.035mm (1.00oz)** | | **Signal** | **GBR** |
| Core | | 1.48mm | FR4_7628 | Dielectric | |
| **Copper** | **L2 (Sig, PWR)** | **0.035mm (1.00oz)** | | **Signal** | **GBR** |
| | B.Mask | 0.02mm | Solder Resist | Solder Mask | GBR |
| | B.Silkscreen | | Direct Printing | Legend | GBR |
| | B.Paste | | | Paste Mask | |

Total thickness: 1.59mm
Note: external layer thicknesses are specified after plating

## Impedance Table

| Transmission Line | Impedance [ohms] | Tolerance [ohms] | Layer | Trace Width [mm] | Gap [mm] | Ref. Layers |
|---|---|---|---|---|---|---|
| Edge-Coupled Coated Microstrip | 100 | ±10 % | L1 | 0.2032 | 0.28 | L2 |

?× Ø??

Ø??

FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

1) FABRICATE PER IPC-6012A CLASS 2.

2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

   DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING FOR REFERENCE ONLY.

3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

   SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.

4) SURFACE FINISH: IMMERSION GOLD

5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR BLUE.

6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.

7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.

8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING

9) PCB MATERIAL REQUIREMENTS:

   A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
   B. Tg 170 C OR EQUIVALENT.
   C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE, AND PROCESS BY A QPL BUILDER FOR SPOTS.

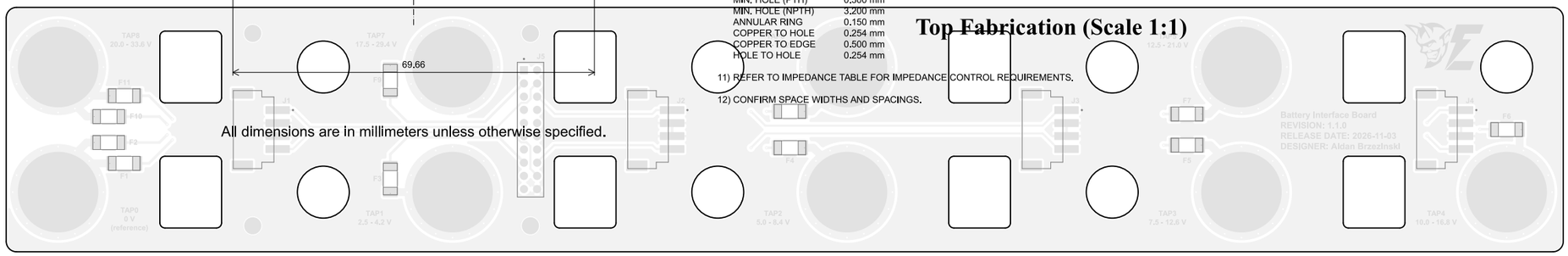
10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

| | |
|---|---|
| BOARD SIZE | 300.000 × 47.000 mm |
| BOARD THICKNESS | 1.590 mm |
| TRACE WIDTH | 0.400 mm |
| TRACE TO TRACE | 0.200 mm |
| MIN. HOLE (PTH) | 0.300 mm |
| MIN. HOLE (NPTH) | 3.200 mm |
| ANNULAR RING | 0.150 mm |
| COPPER TO HOLE | 0.254 mm |
| COPPER TO EDGE | 0.500 mm |
| HOLE TO HOLE | 0.254 mm |

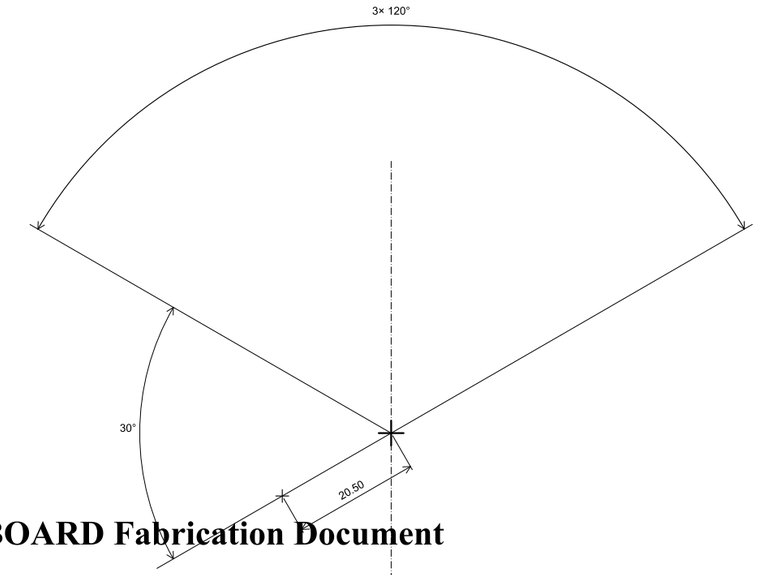
11) REFER TO IMPEDANCE TABLE FOR IMPEDANCE CONTROL REQUIREMENTS.

12) CONFIRM SPACE WIDTHS AND SPACINGS.

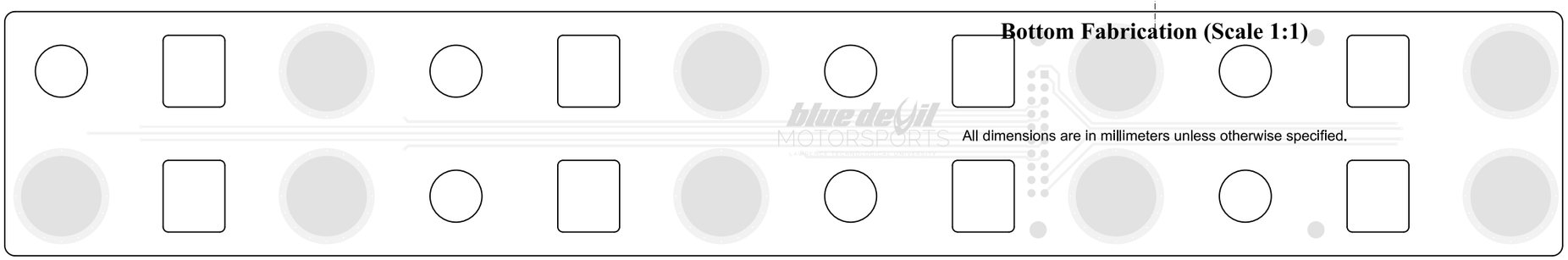
## Top Fabrication (Scale 1:1)




| | | | | |
|---|---|---|---|---|
| Comments: | Company: **Blue Devil Motor Sports** | | Variant: RELEASED | Git Hash: c504242 |
| | Board Name: **BATTERY_INTERFACE_BOARD** | | Project Name: **Battery Interface Board** | |
| Sheet Title: Top Fabrication (Scale 1:1) | File Name: battery_interface_board.kicad_pcb | Designer: Aidan Brzezinski | Date: 2026-10-12 | Revision: 1.1.0 |
| Sheet Path: | | Reviewer: | Size: **A3** | Sheet: **1** of **8** |




# BATTERY_INTERFACE_BOARD Fabrication Document



**Bottom Fabrication (Scale 1:1)**

All dimensions are in millimeters unless otherwise specified.

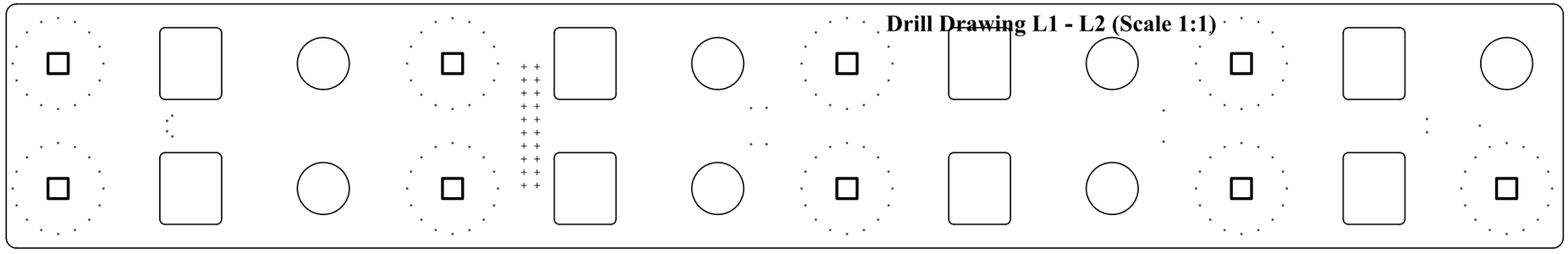
| | | | | | |
|---|---|---|---|---|---|
| Comments: | Company: **Blue Devil Motor Sports** | | Variant: RELEASED | Git Hash: c504242 | |
| | Board Name: **BATTERY_INTERFACE_BOARD** | | Project Name: **Battery Interface Board** | | |
| Sheet Title: Bottom Fabrication (Scale 1:1) | File Name: battery_interface_board.kicad_pcb | Designer: Aidan Brzezinski | Date: 2026-10-12 | Revision: 1.1.0 | |
| Sheet Path: | | Reviewer: | Size: **A3** | Sheet: **2** of **8** | |

## Drill Table

| Symbol | Count | Hole Size | Plated | Hole Shape | Drill Layer Pair | Hole Type |
|---|---|---|---|---|---|---|
| × | 144 | 0.30mm (11.81mils) | PTH | Round | L1 (Sig, PWR) - L2 (Sig, PWR) | Pad |
| ○ | 13 | 0.30mm (11.81mils) | PTH | Round | L1 (Sig, PWR) - L2 (Sig, PWR) | Via |
| + | 20 | 1.02mm (40.16mils) | PTH | Round | L1 (Sig, PWR) - L2 (Sig, PWR) | Pad |
| □ | 9 | 5.50mm (216.54mils) | PTH | Round | L1 (Sig, PWR) - L2 (Sig, PWR) | Pad |
| | Total 186 | | | | | |

# BATTERY_INTERFACE_BOARD Fabrication Document

### Drill Drawing L1 - L2 (Scale 1:1)



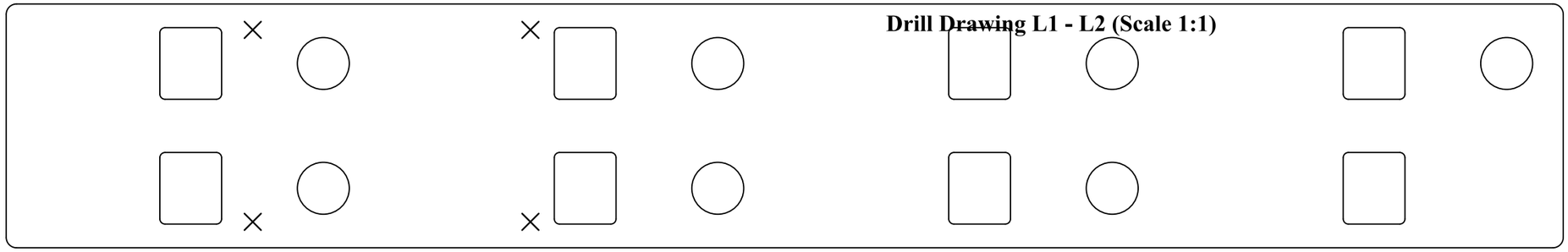
| | | | | | |
|---|---|---|---|---|---|
| Comments: | Company: Blue Devil Motor Sports | | | Variant: RELEASED | Git Hash: c504242 |
| | Board Name: BATTERY_INTERFACE_BOARD | | | Project Name: Battery Interface Board | |
| Sheet Title: Drill Drawing (L1 - L2) | File Name: battery_interface_board.kicad_pcb | Designer: Aidan Brzezinski | | Date: 2026-10-12 | Revision: 1.1.0 |
| Sheet Path: | | Reviewer: | | Size: A3 | Sheet: 3 of 8 |

## Drill Table

| Symbol | Count | Hole Size | Plated | Hole Shape | Drill Layer Pair | Hole Type |
|---|---|---|---|---|---|---|
| × | 4 | 3.20mm (125.98mils) | NPTH | Round | L1 (Sig, PWR) - L2 (Sig, PWR) | Mechanical |
| | Total 4 | | | | | |

# BATTERY_INTERFACE_BOARD Fabrication Document

**Drill Drawing L1 - L2 (Scale 1:1)**

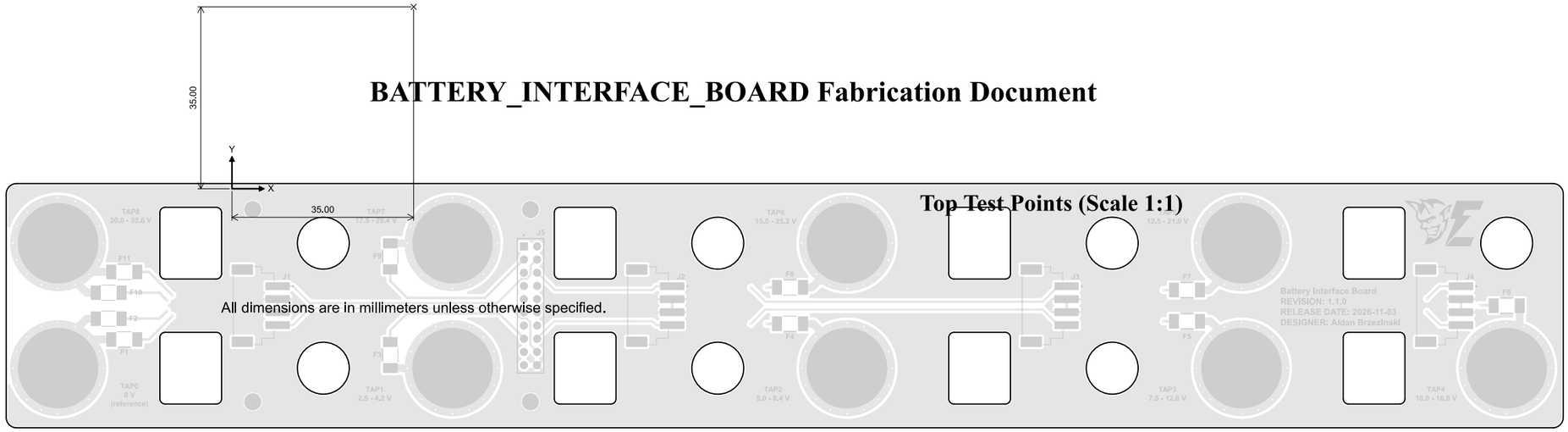


| | | | | | |
|---|---|---|---|---|---|
| Comments: | Company: **Blue Devil Motor Sports** | | | Variant: RELEASED | Git Hash: c504242 |
| | Board Name: **BATTERY_INTERFACE_BOARD** | | | Project Name: **Battery Interface Board** | |
| Sheet Title: Drill Drawing (L1 - L2) | File Name: battery_interface_board.kicad_pcb | Designer: Aidan Brzezinski | | Date: 2026-10-12 | Revision: 1.1.0 |
| Sheet Path: | | Reviewer: | | Size: **A3** | Sheet: **4** of **8** |

| Ref. | Net | X [mm] | Y [mm] |
|---|---|---|---|

| Ref. | Net | X [mm] | Y [mm] |
|---|---|---|---|

# BATTERY_INTERFACE_BOARD Fabrication Document



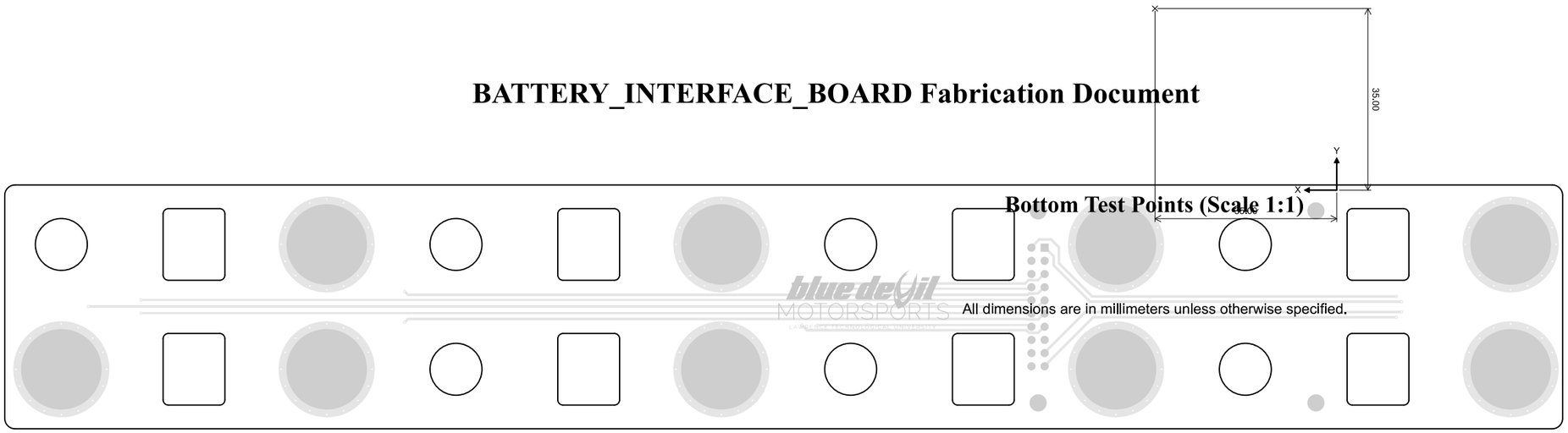

## Top Test Points (Scale 1:1)

All dimensions are in millimeters unless otherwise specified.

| | | | | |
|---|---|---|---|---|
| Comments: | Company: Blue Devil Motor Sports | | Variant: RELEASED | Git Hash: c504242 |
| | Board Name: BATTERY_INTERFACE_BOARD | | Project Name: Battery Interface Board | |
| Sheet Title: Top Test Points (Scale 1:1) | File Name: battery_interface_board.kicad_pcb | Designer: Aidan Brzezinski | Date: 2026-10-12 | Revision: 1.1.0 |
| Sheet Path: | | Reviewer: | Size: A3 | Sheet: 5 of 8 |

| Ref. | Net | X [mm] | Y [mm] |
|---|---|---|---|

# BATTERY_INTERFACE_BOARD Fabrication Document



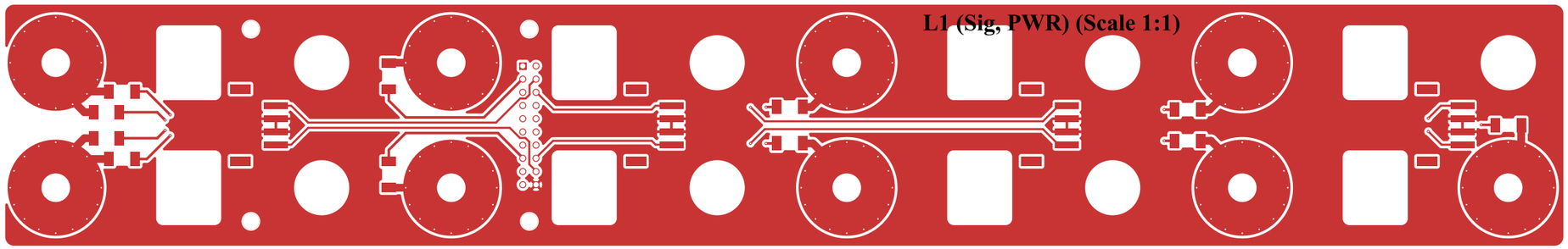

**Bottom Test Points (Scale 1:1)**

All dimensions are in millimeters unless otherwise specified.

| | | | | |
|---|---|---|---|---|
| Comments: | Company: **Blue Devil Motor Sports** | | Variant: RELEASED | Git Hash: c504242 |
| | Board Name: **BATTERY_INTERFACE_BOARD** | | Project Name: **Battery Interface Board** | |
| Sheet Title: Bottom Test Points (Scale 1:1) | File Name: battery_interface_board.kicad_pcb | Designer: Aidan Brzezinski | Date: 2026-10-12 | Revision: 1.1.0 |
| Sheet Path: | | Reviewer: | Size: **A3** | Sheet: **6** of **8** |

# BATTERY_INTERFACE_BOARD Fabrication Document

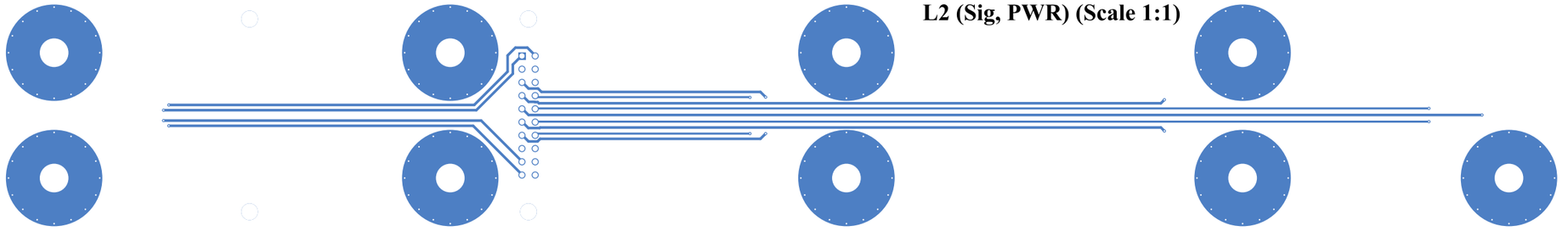
**L1 (Sig, PWR) (Scale 1:1)**



| | | | | |
|---|---|---|---|---|
| Comments: | Company: **Blue Devil Motor Sports** | | Variant: RELEASED | Git Hash: c504242 |
| | Board Name: **BATTERY_INTERFACE_BOARD** | | Project Name: **Battery Interface Board** | |
| Sheet Title: L1 (Sig, PWR) (Scale 1:1) | File Name: battery_interface_board.kicad_pcb | Designer: Aidan Brzezinski | Date: 2026-10-12 | Revision: 1.1.0 |
| Sheet Path: | | Reviewer: | Size: **A3** | Sheet: **7** of **8** |

# BATTERY_INTERFACE_BOARD Fabrication Document

**L2 (Sig, PWR) (Scale 1:1)**



| | | | | |
|---|---|---|---|---|
| Comments: | Company: **Blue Devil Motor Sports** | | Variant: RELEASED | Git Hash: c504242 |
| | Board Name: **BATTERY_INTERFACE_BOARD** | | Project Name: **Battery Interface Board** | |
| Sheet Title: L2 (Sig, PWR) (Scale 1:1) | File Name: battery_interface_board.kicad_pcb | Designer: Aidan Brzezinski | Date: 2026-10-12 | Revision: 1.1.0 |
| Sheet Path: | | Reviewer: | Size: **A3** | Sheet: **8** of **8** |